

ANDROID PROGRAMS
VIVA QUESTIONS & ANSWERS

Mobile Application Development Lab

Program 1: Hello World Application

Q: What is Android?

A: Android is an open-source, Linux-based operating system designed primarily for mobile devices such as smartphones and tablets. It is developed by Google.

Q: What is an Activity in Android?

A: An Activity is a single, focused screen that the user can interact with. It is a core component of Android and serves as the entry point for interacting with the user. Each activity has a lifecycle (onCreate, onStart, onResume, onPause, onStop, onDestroy).

Q: What does the onCreate() method do?

A: onCreate() is called when the activity is first created. It is used to initialize the activity, such as inflating the layout (setContentView), initializing views, and setting up data. It receives a Bundle parameter for restoring previous state.

Q: What is setContentView()?

A: setContentView() sets the activity's layout from an XML resource. It inflates the XML layout file and places it on the screen. For example, setContentView(R.layout.activity_main) loads activity_main.xml.

Q: What is EdgeToEdge.enable(this)?

A: EdgeToEdge enables edge-to-edge display, allowing the app to draw behind system bars (status bar, navigation bar). It was introduced in Android 11 to provide a more immersive experience.

Q: What is the AndroidManifest.xml file?

A: AndroidManifest.xml is a mandatory XML file in every Android app that declares essential information: package name, app components (activities, services, receivers), permissions, minimum API level, and hardware/software features the app requires.

Q: What is the difference between match_parent and wrap_content?

A: match_parent makes the view take up the entire available space of its parent container. wrap_content makes the view only as big as its content (text, image, etc.).

Program 2: Display Message Based on Screen Orientation

Q: What is an Intent in Android?

A: An Intent is a messaging object used to request an action from another app component. It can be used to start an activity, start a service, or deliver a broadcast message. Intents can be explicit (targeting a specific component) or implicit (describing an action for the system to resolve).

Q: What is the difference between Explicit and Implicit Intent?

A: Explicit Intent specifies the target component directly by class name (e.g., `Intent(MainActivity.this, NextActivity.class)`). It is used to start a specific component within the same app. Implicit Intent declares a general action (e.g., `Intent.ACTION_VIEW`) and lets the system find an appropriate component from other apps to handle it.

Q: How does screen orientation work in Android?

A: Screen orientation can be controlled using the `android:screenOrientation` attribute in the manifest. Values include "portrait", "landscape", "sensor", "fullSensor", "user", etc. When orientation changes, Android can restart the activity by default (`onCreate` is called again).

Q: What is the `android:screenOrientation` attribute used for?

A: It locks or specifies the orientation of an activity. For example, `android:screenOrientation="portrait"` forces the activity to stay in portrait mode, while "landscape" forces landscape mode.

Q: What is `startActivity()`?

A: `startActivity()` is a method that launches a new activity. It takes an Intent parameter. For example: `startActivity(new Intent(this, SecondActivity.class))`.

Program 3: Login Window Using UI Controls

Q: What is a `LinearLayout`?

A: `LinearLayout` is a `ViewGroup` that arranges its children in a single direction: either vertically (`android:orientation="vertical"`) or horizontally (`android:orientation="horizontal"`).

Q: What is an `EditText`?

A: `EditText` is a UI widget that allows users to enter and modify text. It supports various input types (text, number, password, email, etc.) via the `android:inputType` attribute.

Q: What is `android:inputType="textPassword"`?

A: It masks the user's input with dots or asterisks so the password is not visible on screen.

Q: What is `android:hint`?

A: `android:hint` displays a suggestion or prompt text inside the `EditText` when it is empty. It disappears as soon as the user starts typing.

Q: What is the purpose of `android:layout_gravity`?

A: `android:layout_gravity` sets the alignment of a view within its parent container. Common values: center, left, right, top, bottom, fill.

Program 4: Explicit and Implicit Intent

Q: How do you pass data between activities using `Intent`?

A: Data is passed using the `putExtra()` method with a key-value pair. For example: `intent.putExtra("key", "value")`. The receiving activity retrieves it with `getIntent().getStringExtra("key")`.

Q: What is `Intent.ACTION_VIEW`?

A: `Intent.ACTION_VIEW` is a standard action that displays data to the user. For example, opening a URL in a browser: `new Intent(Intent.ACTION_VIEW, Uri.parse("http://google.com"))`.

Q: What is `Uri.parse()`?

A: `Uri.parse()` converts a string representation of a URI into a `Uri` object that Android APIs can use. For example: `Uri.parse("http://www.google.com")`.

Q: What is `Intent.createChooser()`?

A: `createChooser()` creates a system dialog that lets the user choose from multiple apps that can handle the given intent. For example: `startActivity(Intent.createChooser(intent, "Choose an app"))`.

Q: Can an implicit intent start an activity in the same app?

A: Yes, if the activity has an intent filter matching the implicit intent. However, implicit intents are primarily used to invoke components from other apps. For same-app navigation, explicit intents are preferred.

Program 5: Custom Designed Opening Screen (Splash)

Q: What is a Splash Screen?

A: A splash screen is a startup screen displayed when an app launches. It typically shows the app logo, branding, or loading animation while the app initializes resources in the background.

Q: What is a Handler in Android?

A: Handler is a class used to schedule tasks on a thread or delay execution. It can post a Runnable to run after a specified delay using `postDelayed()`.

Q: How does `postDelayed()` work?

A: `postDelayed(Runnable r, long delayMillis)` executes the Runnable's `run()` method after the specified delay in milliseconds. For example: `handler.postDelayed(() -> { /* code */ }, 3000)` runs after 3 seconds.

Q: What is the difference between `finish()` and `onDestroy()`?

A: `finish()` is called to close an activity. It triggers the activity lifecycle: `onPause()` → `onStop()` → `onDestroy()`. `onDestroy()` is the final callback before the activity is destroyed.

Q: What is a RelativeLayout?

A: `RelativeLayout` is a `ViewGroup` that positions child views relative to each other or to the parent. Views can be placed using attributes like `android:layout_centerInParent`, `android:layout_below`, `android:layout_toRightOf`, etc.

Q: What is android:background used for?

A: android:background sets the background of a view. It can be a color (@color/...), a drawable (@drawable/...), or a custom XML shape.

Program 6: Show All Views

Q: What is a ScrollView?

A: ScrollView is a layout container that allows its content to scroll vertically. It can hold only one direct child (usually a LinearLayout). When content exceeds the screen height, the user can scroll to see more.

Q: What is android:fillViewport?

A: android:fillViewport="true" makes the ScrollView expand its child to fill the viewport when the child is smaller than the ScrollView.

Q: What is a Spinner and how is it populated?

A: A Spinner is a dropdown selection widget. It is populated by creating an ArrayAdapter with a list of items, then calling spinner.setAdapter(adapter).

Q: What is an ArrayAdapter?

A: ArrayAdapter is a class that bridges data (an array or list) with a view (like a Spinner or ListView). It provides the layout for each item and handles the display.

Q: What is a NumberPicker?

A: NumberPicker is a widget that allows users to select a number from a predefined range by scrolling through values. The range is set with `setMinValue()` and `setMaxValue()`.

Q: What is a ProgressBar?

A: ProgressBar is a visual indicator showing the progress of an operation. It can be indeterminate (circular spinning) or determinate (horizontal bar with a progress value).

Q: What are DatePicker and TimePicker?

A: DatePicker allows users to select a date (year, month, day). **TimePicker** allows users to select a time (hour, minute). Both provide a touch-friendly UI for date/time selection.

Q: What is a RadioGroup and a RadioButton?

A: RadioGroup is a container that holds **RadioButtons**. Only one **RadioButton** within a **RadioGroup** can be selected at a time. It provides mutual exclusion.

Program 7: Create a Menu in Application

Q: How do you create a menu in Android?

A: Create an XML file in the `res/menu/` directory. Use the `<menu>` tag with `<item>` elements for each menu option. Then, in the activity, override `onCreatePanelMenu()` (or `onCreateOptionsMenu()`) to inflate the menu XML using `getMenuInflater()`.

Q: What is MenuInflater?

A: MenuInflater is a class that converts XML menu resources into Menu objects. It is obtained via `getMenuInflater()` and used with `inflate(R.menu.menus, menu)`.

Q: What is a Toast in Android?

A: Toast is a small popup message that appears briefly on the screen and then disappears automatically. It is used for simple feedback and does not block user interaction.

Q: What is onOptionsItemSelected()?

A: This method is called when a menu item is selected. It receives the selected MenuItem. You can read `item.getTitle()` or `item.getItemId()` to identify which item was tapped.

Q: What is the difference between onCreateOptionsMenu() and onCreatePanelMenu()?

A: `onCreateOptionsMenu()` is the traditional method for inflating options menu. `onCreatePanelMenu()` is a newer variant introduced in later Android versions (API 14+) that provides the `featureId` parameter. Both serve the same purpose.

Program 8: Read/Write Local Data (SharedPreferences)

Q: What is SharedPreferences?

A: SharedPreferences is a key-value storage mechanism in Android for storing primitive data (strings, integers, booleans, floats). Data is persisted across app sessions and stored as an XML file in the app's private directory.

Q: How do you write data to SharedPreferences?

A: Call `getSharedPreferences("pref_name", MODE_PRIVATE)` to get the preferences object, then `edit()` to get an Editor. Use `putString()`, `putInt()`, etc. to store values, and call `apply()` or `commit()` to save.

Q: What is the difference between `apply()` and `commit()` in SharedPreferences?

A: `apply()` saves changes asynchronously (in the background) and does not return a result. `commit()` saves synchronously and returns a boolean indicating success. `apply()` is preferred for most cases as it does not block the UI thread.

Q: How do you read data from SharedPreferences?

A: Use `getString(key, defaultValue)`, `getInt(key, defaultValue)`, etc. on the SharedPreferences object. The default value is returned if the key does not exist.

Q: What is the `onResume()` method used for in this context?

A: `onResume()` is called when the activity becomes visible and interactive. In the `SharedPreferences` example, `onResume()` is used to reload saved data when the app returns to the foreground, ensuring the fields show the latest values.

Q: What is `Context.MODE_PRIVATE`?

A: `MODE_PRIVATE` means the `SharedPreferences` file can only be accessed by the creating app. Other apps cannot read it. This is the standard security mode for app-private data.

Program 9: Create/Read/Write Data with SQLite Database

Q: What is SQLite?

A: SQLite is a lightweight, embedded relational database engine that comes built-in with Android. It stores data in a single file on the device and supports standard SQL queries. It requires no separate server process.

Q: What is `openOrCreateDatabase()`?

A: It creates a new SQLite database file if it does not exist, or opens an existing one. Parameters: database name, mode (usually `MODE_PRIVATE`), and a `CursorFactory` (usually null). Returns a `SQLiteDatabase` object.

Q: What is SQL injection and how does it apply here?

A: SQL injection is a security vulnerability where an attacker inserts malicious SQL code through user inputs. In this program, user input is directly concatenated into SQL queries (e.g., "INSERT INTO student VALUES('" + Rollno.getText() + "')"), which is vulnerable. The safer approach is to use parameterized queries with ? placeholders.

Q: What is a Cursor in Android?

A: A Cursor is an interface that provides read access to the result set of a database query. It allows iterating through rows (moveToFirst(), moveToNext()) and reading column values (getString(index), getInt(index)).

Q: What is db.execSQL() used for?

A: execSQL() executes a SQL statement that does not return a result set (e.g., INSERT, UPDATE, DELETE, CREATE TABLE). For queries that return data (SELECT), use.rawQuery() instead.

Q: What is the purpose of AlertDialog.Builder?

A: AlertDialog.Builder is used to create dialog boxes in Android. It allows setting the title, message, buttons, and other properties, then calling show() to display the dialog.

Q: What are CRUD operations?

A: CRUD stands for Create (INSERT), Read (SELECT), Update (UPDATE), and Delete (DELETE). These are the four basic operations for persistent storage.

Program 10: Send SMS

Q: What is SmsManager?

A: SmsManager is a system service class that manages SMS operations. The static getDefault() method returns the default SmsManager instance. It provides sendTextMessage() to send SMS messages.

Q: What permissions are required to send SMS?

A: The SEND_SMS permission is required. On Android 6.0+ (API 23), dangerous permissions must be requested at runtime using ActivityCompat.requestPermissions().

Q: What is runtime permission in Android?

A: Starting from Android 6.0, dangerous permissions (e.g., SMS, Camera, Location) must be requested while the app is running, not just declared in the manifest. The app shows a dialog to the user, who can grant or deny the permission.

Q: What is ContextCompat.checkSelfPermission()?

A: It checks whether the app has been granted a particular permission. Returns PackageManager.PERMISSION_GRANTED or PERMISSION_DENIED.

Q: What is the <uses-feature> tag in the manifest?

A: `<uses-feature>` declares a hardware or software feature that the app requires. `android:required="false"` means the app can function without the feature, but will use it if available.

Program 11: Send Email

Q: How do you send an email from an Android app?

A: Use an implicit Intent with `Intent.ACTION_SEND`, `setType("text/plain")`, and add extras: `EXTRA_EMAIL` (recipient), `EXTRA_SUBJECT` (subject), and `EXTRA_TEXT` (body). Then call `startActivity(Intent.createChooser(intent, "Choose app"))`.

Q: What is `Intent.EXTRA_EMAIL`?

A: `EXTRA_EMAIL` is a string array that holds the recipient email addresses. It is passed as an extra in the email intent.

Q: Why is `Intent.createChooser()` used?

A: `createChooser()` ensures the user sees a dialog to pick their preferred email app. Without it, the system might auto-launch a default app or crash if no suitable app is found.

Q: What does `intent.resolveActivity()` do?

A: `resolveActivity()` checks if there is at least one app on the device that can handle the given intent. Returns null if no matching app is found.

Q: What is the purpose of `setType("text/plain")`?

A: setType() sets the MIME type of the data being sent. "text/plain" indicates plain text. For emails with attachments, you might use "message/rfc822".

Program 12: Display Map Based on Current/Given Location

Q: What is Google Maps Android API?

A: The Google Maps Android API allows developers to embed Google Maps into Android apps. It provides features like map display, markers, camera control, location tracking, and custom overlays.

Q: What is SupportMapFragment?

A: SupportMapFragment is a fragment that hosts a Google Map in the app's layout. It is part of the Google Play Services library and provides backwards compatibility for older API versions.

Q: What is the OnMapReadyCallback interface?

A: OnMapReadyCallback is called when the map is ready to be used. The onMapReady(GoogleMap map) method provides a GoogleMap object that can be configured (add markers, move camera, etc.).

Q: What is CameraUpdateFactory?

A: CameraUpdateFactory is used to create camera update objects that control the map's viewport. newLatLngZoom(LatLng, zoom) creates an update that moves the camera to a specific location at a given zoom level.

Q: What is a LatLng?

A: LatLng is an immutable data class representing a geographic coordinate (latitude and longitude). It is used to specify locations on the map.

Q: What is required to use Google Maps in an app?

A: You need: (1) A Google Maps API key added to the manifest as a `<meta-data>` element, (2) Google Play Services SDK in build.gradle, (3) Permissions (`ACCESS_FINE_LOCATION`, `INTERNET`), and (4) A `SupportMapFragment` or `MapView` in the layout.

Program 13: Login Module with Validation

Q: What is input validation in Android?

A: Input validation ensures that user-entered data is correct and complete before processing. For login: checking if username/password fields are empty, verifying credentials match expected values, and providing feedback via Toast or TextView.

Q: How do you check if an EditText is empty?

A: Call `getText().toString().isEmpty()` (or `.length() == 0`). If true, the field is empty and you can display an error message.

Q: What is the difference between Toast and setText() for showing messages?

A: Toast appears temporarily on screen and auto-dismisses. `setText()` on a `TextView` changes text permanently (until changed again). In the login app, success is shown via `setText()` and errors via Toast.

Q: What is hardcoded credential checking?

A: The example compares username and password against hardcoded strings "Admin"/"Admin". In real apps, credentials should be hashed and verified against a backend server or database.

Q: What is `findViewById()`?

A: `findViewById()` is a method that locates a view by its XML ID and returns the corresponding object. For example: `EditText et = (EditText) findViewById(R.id.etUserName)`.

Program 14: Deploy Android Application

Q: What is an APK file?

A: APK (Android Package Kit) is the file format used to distribute and install Android apps. It contains the compiled code (DEX), resources, manifest, and certificates.

Q: What is a keystore in Android?

A: A keystore is a binary file containing private keys used to sign an Android app. It ensures app integrity and establishes trust. The same keystore must be used for app updates to prove ownership.

Q: What is the difference between debug APK and release APK?

A: Debug APK is signed with a default debug keystore (for testing) and is not suitable for distribution. Release APK is signed with a custom production keystore and is optimized (ProGuard/R8).

Q: What is the Google Play Console?

A: Google Play Console is a web platform where developers manage Android app releases: upload APK/AAB, set pricing, view analytics, respond to reviews, and manage beta testing.

Q: What is an AAB (Android App Bundle)?

A: AAB is the modern publishing format replacing APK on Google Play. It contains all compiled code and resources but defers APK generation and signing to Google Play, enabling optimized downloads (smaller APKs per device).

Q: What are the steps to publish an app on Google Play?

A: (1) Create a developer account, (2) Prepare the app (debugging off, versioned), (3) Generate a signed release AAB/APK, (4) Create a store listing (title, description, screenshots), (5) Upload the AAB/APK, (6) Set pricing, (7) Publish.

General Android Questions

Q: What are the four main components of Android?

A: (1) Activity — UI screen, (2) Service — background processing, (3) BroadcastReceiver — responds to system-wide events, (4) ContentProvider — manages shared app data.

Q: What is the Android Activity Lifecycle?

A: The lifecycle consists of: onCreate() → onStart() → onResume() → onPause() → onStop() → onDestroy(). onRestart() is called between onStop() and onStart().

Q: What is the difference between padding and margin?

A: Padding is the space inside the view's border (between border and content). Margin is the space outside the view's border (between this view and adjacent views).

Q: What is an XML layout in Android?

A: XML layouts are declarative UI definitions stored in res/layout/. They separate UI design from code logic. Android uses XML namespaces (android:, app:, tools:) to set attributes.

Q: What is Gradle in Android?

A: Gradle is the build system used by Android Studio. It automates compilation, packaging, signing, and dependency management. The build.gradle file configures SDK versions, libraries, and build types.

Q: What is R.java?

A: R.java is an auto-generated file that contains resource IDs for all resources in the res/ directory (layouts, strings, drawables, etc.). It allows code to reference resources like R.layout.activity_main or R.id.myButton.

Q: What is the difference between Serializable and Parcelable?

A: Both are used to pass objects via Intent. Serializable is a Java-standard interface that uses reflection (slower). Parcelable is an Android-specific interface that is much faster and recommended for Android development.

Q: What is ProGuard/R8?

A: ProGuard and R8 are tools that shrink, obfuscate, and optimize the compiled bytecode. They remove unused code and rename classes/methods to make reverse engineering harder and reduce APK size.

Q: What is a Fragment?

A: A Fragment is a reusable portion of an Activity's UI. Fragments have their own lifecycle and can be combined to create multi-pane layouts (e.g., tablets). They are hosted by an Activity.

Q: What is the constraint layout?

A: ConstraintLayout is a flexible layout that positions views using constraints (connections to parent, other views, or guidelines). It creates flat view hierarchies and is the default layout in Android Studio templates.